Chuheng Wei

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RESEARCH INTERESTS

- Autonomous Driving under Non-Ideal Environmental and Sensing Conditions
- Cooperative Autonomous Systems Integrating Perception and Control
- Human-Centered and Personalized Autonomous Intelligence
- End-to-End Autonomous Driving and Vision-Language-Action (VLA) for Embodied Mobility Systems

EDUCATION

Ph.D. University of California, Riverside — **Riverside, US** — 09/2022 – 05/2026

- Electrical and Electronics Engineering
- Advisors: Dr. Matthew Barth and Dr. Guoyuan Wu
- Dissertation Topic: *Cooperative Perception under Adverse Conditions: Addressing Challenges and Enhancing Performance*

M.Sc. University of Warwick — **Coventry, UK** — 10/2021 – 10/2022

- Smart, Connected and Autonomous Vehicles
- Advisors: Dr. Valentina Donzella
- Dissertation: *Raw camera data object detectors: an optimisation for automotive video processing and transmission*

B.Eng. Communication University of China — **Beijing, China** — 09/2013 – 07/2017

- Digital Media Technology
- Dissertation: *Design and Implementation of a Machine Learning–Based Vision Target Detection System for Mobile Shooting Robots*

HONORS & AWARDS

Best Paper Nomination — IEEE ITSC 2025	2025
1st Place – 2025 CVPR V2X Cooperative Perception Challenge	2025
Esther F. Hays Graduate Award, UC Riverside	2025
Award – USDOT Intersection Safety Challenge (Stage 1B) Prize	2024
Best Student Poster Runner-up Award – DSAA 2024	2024
Esther F. Hays Graduate Award, UC Riverside	2024
Third Prize — RoboMasters National Finals	2017
Second Prize — North China Robot Competition	2016
Third Prize — China Robot Competition, CAA	2016
Third Prize — RoboMasters North Region	2016
Second Prize — Beijing College Student Robot Competition	2016

PUBLICATIONS

Book

V2X-Enabled Cooperative Driving Systems — Proposal Accepted
G. Wu, ***C. Wei***
Book proposal accepted by Springer Nature; full manuscript in preparation

Journal Articles

[J5] **DOV-VeRi: A Real-World Vehicle Re-identification Dataset for Cross-View Scenarios with Opposing Camera Perspectives** — Under Review
C. Wei, Z. Qin, G. Wu, M. J. Barth, Y. Liu, E. A. Sisbot, K. Oguchi
Under review at IEEE Transactions on Intelligent Transportation Systems (T-ITS)

[J4] **Cooperative Perception for Automated Driving: A Comprehensive Survey of Algorithms, Applications, and Future Directions** — 2025

C. Wei, G. Wu, M. J. Barth
Proceedings of the IEEE (PIEE)

[J3] **Investigating Personalized Driving Behaviors in Dilemma Zones: Analysis and Prediction of Stop-or-Go Decisions** 2025
Z. Qin, S. Li, ***C. Wei***, G. Wu, M. J. Barth, A. Abdelraouf, R. Gupta, K. Han
IEEE Robotics and Automation Letters (RA-L)

[J2] **Raw Camera Data Object Detectors: An Optimisation for Automotive Processing and Transmission** 2025
C. Wei, P. H. Chan, A. Huggett, V. Donzella
IEEE Access

[J1] **Vulnerable Road User Detection for Roadside-Assisted Safety Protection: A Comprehensive Survey** 2025
Z. Zhang, ***C. Wei***, G. Wu, M. J. Barth
Applied Sciences

Conference Papers

[C20] **PDB: Not All Drivers Are the Same – A Personalized Dataset for Understanding Driving Behavior** Under Review
C. Wei, Z. Qin, S. Li, Z. Zhang, X. Zhao, G. Wu, M. J. Barth, A. Abdelraouf, R. Gupta, K. Han
Arxiv

[C20] **ViLi-Speed: A Real-Time Lightweight Computer Vision Approach for Virtual Line-Based Speed Estimation** Accepted
C. Wei, Z. Qin, K. BoriboonsomsinG. Wu
Arxiv

[C18] **Precision Camera Calibration for AI-Powered Traffic Surveillance at Urban Intersections** Accepted
C. Wei, Z. Zhao, M. Han, K. Boriboonsomsin, G. Wu
7th International Conference on Artificial Intelligence, Robotics, and Control (IEEE AIRC),

[C17] **Beyond Perception: A Survey and Future Directions of VLM-based Autonomous Driving Datasets** Accepted
C. Wei, H. Cheng, Z. Zhang, G. Wu,Z. Wang M. J. Barth
IEEE Intelligent Vehicles Symposium (IV 2026)

[C16] **HeCoFuse: Cross-Modal Complementary V2X Cooperative Perception with Heterogeneous Sensors** 2025
C. Wei, Z. Qin, W. Zimmer, G. Wu, M. J. Barth
IEEE 28th International Conference on Intelligent Transportation Systems (ITSC 2025)

[C15] **PointCompress3D: A Point Cloud Compression Framework for Roadside LiDARs in Intelligent Transportation Systems** 2025
W. Zimmer, R. Pranamulia, X. Zhou, J. Zhang, ***C. Wei***, R. Greer, J. S. B. Perez, M. Liu, M. M. Trivedi, A. Knoll
IEEE 28th International Conference on Intelligent Transportation Systems (ITSC 2025)

[C14] **Hierarchical Multi-Modal Fusion for Roadside VRU Detection: Method Complementarity Under Sparse Label Constraints** 2025
C. Wei, Z. Zhang, H. Liu, G. Wu, M. J. Barth
ICCV 2025, AI-City Workshop

[C13] **EKI-GAN: Context-Aware Vehicle Trajectory Forecasting with Vehicle Factors and Environmental Information at Signalized Intersections** 2025
C. Wei, G. Wu, M. J. Barth
ICCV 2025, AI-City Workshop

[C12] **Contextual-Personalized Adaptive Cruise Control via Fine-Tuned Large Language Models** 2025
Z. Qin, X. Yao, ***C. Wei***, A. Ji, G. Wu, Z. Sun
ICCV 2025, DriveX Workshop

[C11] **Integrating Multi-Modal Sensors: A Review of Fusion Techniques for Intelligent Vehicles** 2025
C. Wei, Z. Qin, Z. Zhang, G. Wu, M. J. Barth
IEEE Intelligent Vehicles Symposium (IV 2025)

[C10] **RAF-RCNN: Adaptive Feature Transfer from Clear to Rainy Conditions for Improved Object Detection** 2024
C. Wei, Z. Qin, G. Wu, M. J. Barth, Z. Sun
CICTP 2024

[C9] **Cost-Efficient Driver Distraction Detection System: Transformer-Based Classification on Bayer Image** 2024
C. Wei, G. Wu, M. Barth
IEEE 27th International Conference on Intelligent Transportation Systems (ITSC 2024)

[C8] **Dilemma Zone: A Comprehensive Study of Influential Factors and Behavior Analysis** 2024
C. Wei, Z. Qin, G. Wu, M. J. Barth, A. Abdelraouf, R. Gupta, K. Han
2024 Forum for Innovative Sustainable Transportation Systems (FISTS)

[C7] **Feature Corrective Transfer Learning: End-to-End Solutions to Object Detection in Non-Ideal Visual Conditions** 2024
C. Wei, G. Wu, M. J. Barth
CVPR 2024, UG2+ Workshop

[C6] **KI-GAN: Knowledge-Informed Generative Adversarial Networks for Enhanced Multi-Vehicle Trajectory Forecasting at Signalized Intersections** 2024
C. Wei, G. Wu, M. J. Barth, A. Abdelraouf, R. Gupta, K. Han
CVPR 2024, AI City Workshop

[C5] **Enhanced Object Detection by Integrating Camera Parameters into Raw Image-Based Faster R-CNN** 2023
C. Wei, G. Wu, M. J. Barth, P. H. Chan, V. Donzella, A. Huggett
IEEE 26th International Conference on Intelligent Transportation Systems (ITSC 2023)

[C4] **Personalized Trajectory Prediction for Driving Behavior Modeling in Ramp-Merging Scenarios** 2023
S. Li, ***C. Wei***, G. Wu, et al.
IEEE International Conference on Robotic Computing (IRC 2023)

[C3] **Vehicle Detecting and Tracking Application Based on YOLOv5 and DeepSORT for Bayer Data** 2022
C. Wei
2022 17th International Conference on Control, Automation, Robotics and Vision (ICARCV)

[C2] **An Analysis of the Brake Performance of Radar-Based Adaptive Cruise Control During Ramp Merging on Simulation Software** 2022
Z. Zhao, ***C. Wei***
2022 7th International Conference on Intelligent Computing and Signal Processing (ICSP)

[C1] **A Driver Distraction Detection Method Based on Convolutional Neural Network** 2022
C. Wei, C. Liu, S. Chi
2022 7th International Conference on Computer and Communication Systems (ICCCS)

EMPLOYMENT

Amazon,Sunnyvale, CA, US 09/2025 – 12/2025
- Applied Scientist Intern

UC Riverside,Riverside, CA, US 01/2023 – 05/2026
- Research Assistant

Onsemi, Coventry, UK 11/2021 – 09/2022
- Computer Vision Research Assistant

Smart Walnut, Beijing, China 02/2020 – 06/2021
- C++ Teaching & Research Engineer

Bite Zhiyuan Technology Education Co., Beijing, China 07/2017 – 01/2020
- Robotics & Programming Instructor and Competition Coach